

Mitsue Project — Revenue & Payback Summary

One-Page Funder-Facing Overview

All figures are illustrative early-stage estimates. Phase 1 feasibility studies (due M9, Dec 2026) will replace these with vetted figures anchored to real survey data.
Full detail in [mitsue_implementation_plan.md](#) — "Return on Investment" section.

1. Revenue Streams

Biomass CHP (fuelled by sugi forest thinnings) is the **primary energy source**, providing 24/7 baseload electricity and heat. Solar, battery storage, and EV charging are complementary. The circular local energy economy — where the forest powers the village — underpins all revenue streams.

Revenue / Savings Category	Year 1	Year 5	Year 10	Notes
Data center hosting fees	¥0	¥15–30M	¥40–80M	Edge computing / AI specialist workloads
Electricity sales (FIT/FIP)	¥0	¥5–15M	¥10–30M	Surplus from biomass CHP + solar generation
EV charging fees	¥0	¥1–3M	¥3–8M	Growing as EV fleet expands
Carbon credits (J-Credit)	¥0	¥1–3M	¥3–10M	From native forest restoration
Forestry products (timber, lumber)	¥0	¥3–8M	¥10–20M	Beyond fuel-residue use
Education / consulting ("playbook")	¥0	¥1–3M	¥3–8M	Replication licensing and training
Total annual revenue (illustrative)	¥0	¥28–67M	¥74–166M	

2. Cost Displacement

The project *replaces or avoids* significant existing costs:

- Energy imports:** Mitsue residents and businesses currently import ~100% of electricity. Local generation displaces approximately **¥40–60M/yr** of energy spending leaving the village economy.
- Koryukan maintenance burden:** The Koryukan costs the village an estimated **¥3–8M/yr** in basic upkeep with no return. Active reuse as a data center hub converts a stranded asset into a productive facility.
- Forest management liability:** Untended sugi plantations impose ecological costs (pollen, biodiversity loss) and physical risks (landslide, fire). Active management converts this liability into feedstock and timber revenue.

3. Payback Framework

Metric	Value
Total capital deployed (Phases 1–3) by Year 5	¥200–290M
Annual revenue by Year 5	¥28–67M
Operating costs by Year 5 (est. 60% of revenue)	¥18–35M
Net annual surplus by Year 5	¥10–32M
Approximate capital payback period	10–18 years (depending on grant vs. loan vs. revenue-financing mix)
Break-even / net-surplus year (base case)	~Year 5

Once net surplus is achieved, reinvestment priority: (1) forestry scale-up → (2) data-center expansion → (3) replication support for other villages.

4. Honest Caveat

These are **framework figures, not a forecast**. They assume successful execution of all three project elements (biomass CHP energy, data center, forestry) at target scale. Phase 1 feasibility studies will sharpen these numbers significantly. Real-world performance — particularly biomass CHP sizing, data center occupancy, and FIT/FIP tariff outcomes — will determine actual ROI.

Sources

These are the project's **own illustrative early-stage estimates**, consolidated from the "Return on Investment" section of [mitsue implementation plan.md](#) — not externally audited figures. The underlying load, PUE, and carbon assumptions and their references (HIGHRESO PUE target, J-Credit methodology, Forestry Agency sequestration data, edge-data-center power benchmarks) are documented in that plan's Phase 1 "Sources & assumptions" block.

The categories above will be validated against METI FIT/FIP tariff schedules, J-Credit pricing, and biomass/timber market data **during** the Phase 1 feasibility studies (M9, December 2026), which will replace these framework figures with vetted estimates.

Rob Oudendijk — YR-Design / Safecast Mitsue, Nara Prefecture, Japan June 2026